

AI-Powered Supplier Risk & Performance Analytics Dashboard

Dataset: 1,800 records | 100 suppliers | 2021-2025

TOTAL SUPPLIERS

100

TOTAL RECORDS

1,800

HIGH RISK RECORDS

884

49.1%

MEDIUM RISK RECORDS

916

TOTAL INCIDENTS

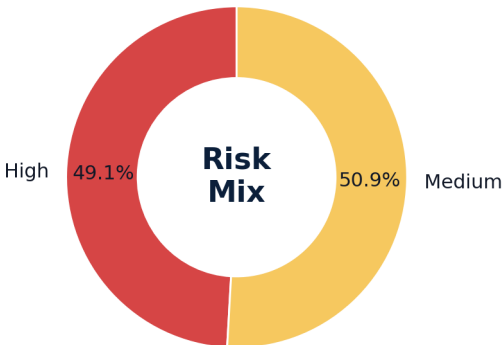
17,282

AVG MCDM SCORE

0.703

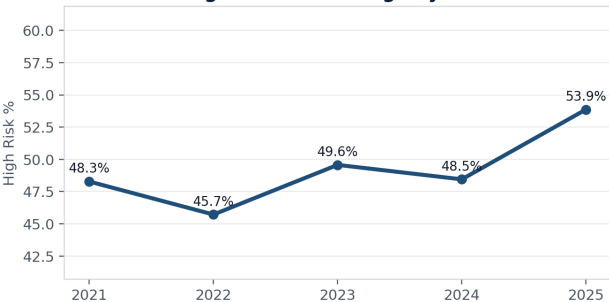
Risk Mix

Risk Category Distribution



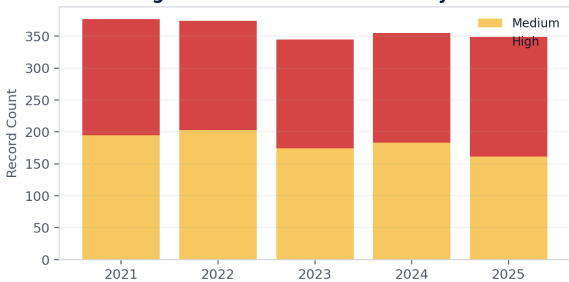
Year-wise High Risk Trend

High Risk Percentage by Year



High vs Medium Records

High vs Medium Risk Records by Year



Executive Business Insights

- High-risk exposure is 49.1% of all supplier records, indicating a material supply continuity risk.
- 2025 has the highest recent high-risk share at 53.9%, requiring immediate supplier review.
- Risk scoring is mainly driven by financial stability, delivery performance and quality compliance.
- Recommended action: create supplier watchlist, diversify backup vendors and prioritize audits for high-risk suppliers.

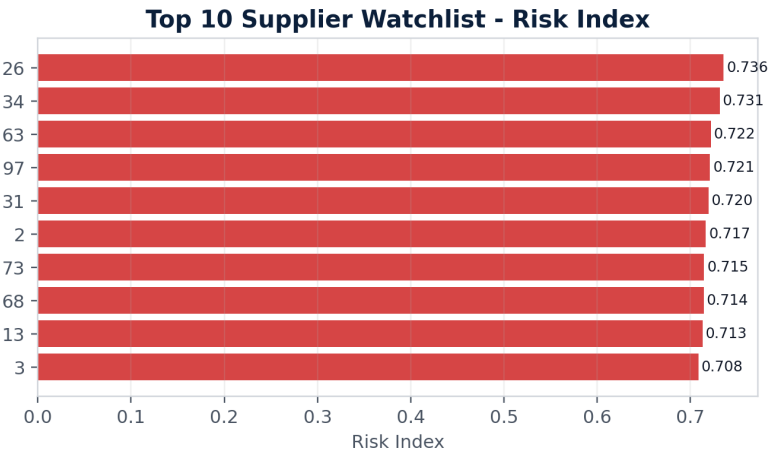
Recommended Management Action

- Review top risk suppliers monthly.
- Reduce dependency on repeated high-risk suppliers.
- Track incident-based risk separately from MCDM score.
- Use AI probability output to prioritize audits.

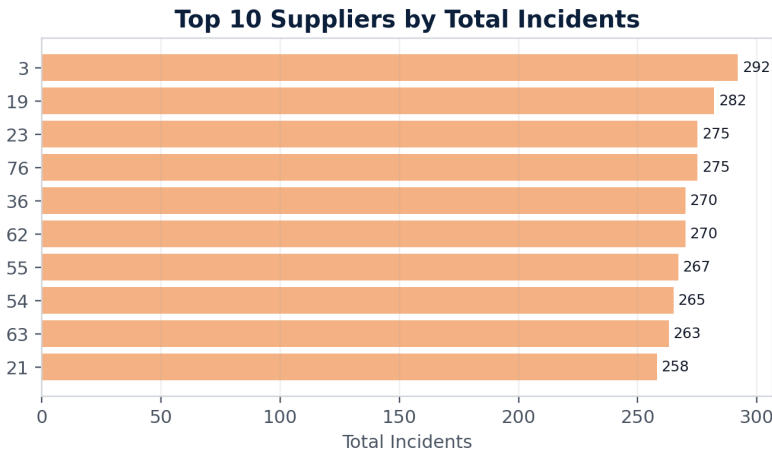
Supplier Watchlist & Incident Dashboard

Focus: high-risk suppliers requiring procurement action

Top Watchlist by Risk Index



Top Suppliers by Total Incidents



Supplier	Avg MCDM	High Risk %	Incidents	Avg Delivery	Avg Quality	Risk Index
26	0.748	77.8%	176	0.71	0.77	0.736
34	0.754	71.4%	144	0.79	0.76	0.731
63	0.717	63.6%	263	0.73	0.70	0.722
97	0.754	68.8%	161	0.79	0.72	0.721
31	0.712	45.5%	161	0.78	0.74	0.720
2	0.719	73.7%	192	0.72	0.77	0.717
73	0.736	72.7%	106	0.73	0.74	0.715
68	0.730	60.0%	233	0.71	0.81	0.714
13	0.729	62.5%	180	0.74	0.80	0.713
3	0.710	58.3%	292	0.73	0.78	0.708

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Supplier Performance Drivers

Focus: what drives supplier risk

TOP AI RISK DRIVER

Financial Stability

37.6% importance

MODEL ACCURACY

93.06%

ROC-AUC

98.83%

PRECISION

93.68%

RECALL

92.09%

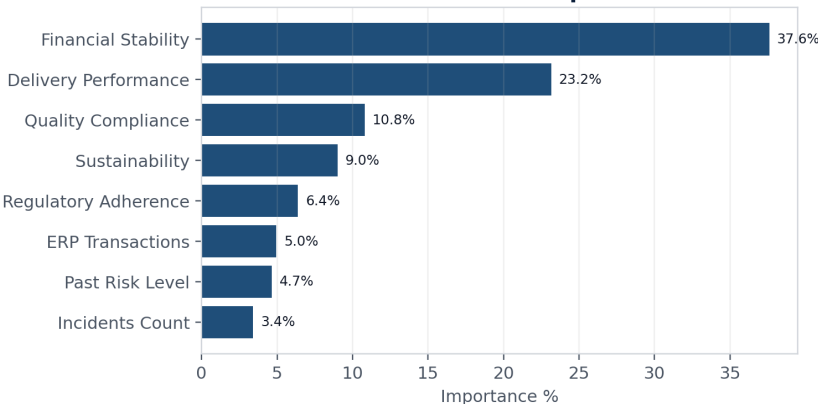
Performance Score Comparison

Average Performance Scores by Risk Category



Feature Importance from AI Model

AI Model Feature Importance



Data Science Interpretation

- Financial Stability has the strongest contribution to predicted supplier risk.
- Delivery Performance is the second strongest driver and should be tracked in procurement reviews.
- Quality Compliance remains a critical operational control metric.
- Past risk, ERP transaction count and incidents can be used for an improved future-risk target.

Procurement Recommendations

- Create tiered supplier segmentation: Critical Watch, High Watch, Medium Monitoring, Low Monitoring.
- Prioritize audits for suppliers with high risk index and high incidents.
- Build fallback supplier network for critical materials.
- Move from reactive vendor review to predictive supplier governance.

AI Risk Prediction Dashboard

Focus: model output and action plan

ACCURACY
93.06%

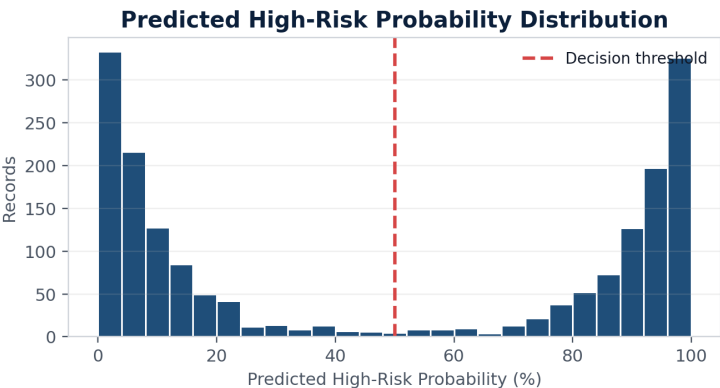
ROC-AUC
98.83%

F1 SCORE
92.88%

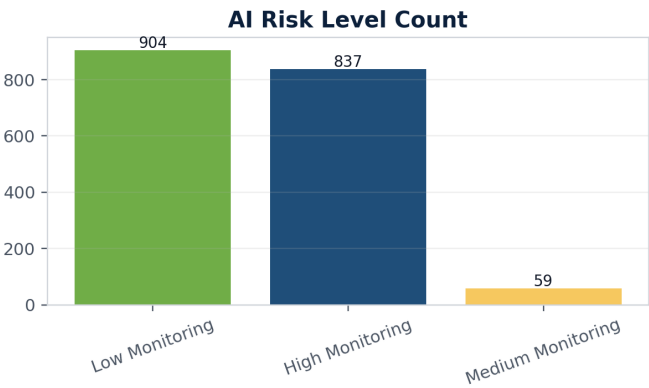
PREDICTED HIGH RISK
882

CRITICAL WATCH
0

Prediction Probability Distribution



AI Risk Level Count



Top 4 Highest AI-Risk Records

Supplier	Year	Actual Risk	Predicted Risk	High Risk Prob	AI Level	Incidents
4	2021	High	High	100.0%	High Monitoring	11
48	2024	High	High	100.0%	High Monitoring	5
52	2023	High	High	100.0%	High Monitoring	6
93	2021	High	High	100.0%	High Monitoring	7

Final Business Summary

This supplier analytics dashboard analyzes 1,800 supplier records across 100 suppliers from 2021-2025. High-risk records represent 49.1% of the dataset. The highest high-risk year is 2025 at 53.9%. The AI model identifies financial stability and delivery performance as the strongest risk drivers. Management should combine MCDM score, incident volume and AI probability to prioritize supplier audits and backup sourcing.